「Def. Let F be a field. Given a polynomial $f(x) \in F[x]$ over F, the Extension

$$E \overset{\text{def}}{=} F(\alpha_1, \cdots, \alpha_n) \quad \text{where } \alpha_1, \cdots, \alpha_n \in \overline{F} \text{ are roots of } f.$$

is called the splitting field of $f(x)$.
Further, the collection of polynomials $\{f_i(x) \mid i \in I\}$, the intersection of splitting fields of f_i is also called the splitting field.

「Thm. Let $\varphi : F \to F'$ be an isomorphism of fields.
Given $f(x) = a_n x^n + \cdots + a_1 x + a_0 \in F[x]$, let $f'(x) = \varphi(a_n)x^n + \cdots + \varphi(a_0) \in F'[x]$.
Let E be the splitting field for $f(x)$ over F,
and E' be the splitting field for $f'(x)$ over F'.
Then, φ can be extended on E to E', isomorphically.

[Dummit. Sec 13.4.

[Prop. Let K be a finite extension of F.

K/F is splitting field

if and only if

every irreducible polynomial in $F[x]$ that has a root in K splits over K.

Proof. Suppose that K is the splitting field of $\{f_i(x) \mid i \in \{1,2,\cdots,n\}, f_i \in F[x]\}$.
Since K/F is finite, the collection $\{f_i\}$ was chosen finite. Simply, we say that K is a splitting field of $f = f_1 \cdots f_n \in F[x]$.] ← 딱 필요한 과정은 아님.

Let $p(x) \in F[x]$ be an irreducible polynomial such that $p(\alpha)=0$ for some $\alpha \in K$.
Then, for any other root $\beta \in \overline{F}$ of $p(x)$, there is a conjugate isomorphism $\psi : F(\alpha) \to F(\beta)$ which fixes F and maps α to β.
Now, by the Extension Theorem, there is an Extension of ψ, $\overline{\psi} : K \to \overline{F}$ such that $\overline{\psi}|_{F(\alpha)} = \psi$. But, since K/F is splitting field and $\overline{\psi}$ fixes F, so $\overline{\psi}[K] = K$. Therefore, $F(\beta) \subseteq K$, so $\beta \in K$.

Conversely, suppose that the below. Since K/F is finite, we can write $K = F(\alpha_1, \cdots, \alpha_r)$. Now, let $f_i \in F[x]$ be a minimal polynomial of α_i.
Let L be the splitting field of $f_1, \cdots, f_r$ over F. By the assumption, all f_i split over K, so $L \subseteq K$. Reversely, each α_i is a root of f_i, so $K = F(\alpha_1, \cdots, \alpha_r) \subseteq L$. Consequently, $K = L$ ☒

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Let $f(x) \in F[x]$ be an irreducible polynomial such that for some $\alpha \in K$, $f(\alpha)=0$.
If $\alpha \in F$, then f is just degree 1 polynomial, so there is nothing to prove.
If $\alpha \notin K$, then α is a root of for some f_j, and since f_j splits over K,
So the irreducible f is also minimal polynomial of α, so f divides f_j, therefore f splits over K.

α가 어떤 f_j의 근일 필요가 없다!

Ex) $\mathbb{Q}(\sqrt{2}, \sqrt{3})$은 $(x^2-2)(x^2-3)$의 splitting field.
하지만 $\sqrt{2}+\sqrt{3}$의 minimal polynomial은 $x^4 - 10x^2 + 1$.

Prop. Let K_1, K_2 be finite extensions of F contained in the field K, and suppose that K_1 and K_2 both are splitting fields over F. Then, K_1K_2 and $K_1 \cap K_2$ both are splitting fields over F.

Proof. Let K_1 and K_2 be splitting field of f_1 and f_2, respectively. Then, for all roots $\alpha_1, \cdots, \alpha_k$ and $\beta_1, \cdots, \beta_r$ of f_1 and f_2, respectively we can write $K_1 = F(\alpha_1, \cdots, \alpha_k)$ and $K_2 = F(\beta_1, \cdots, \beta_r)$
Then, the Composition field $K_1K_2 = F(\alpha_1, \cdots, \alpha_k, \beta_1, \cdots, \beta_r)$.
Meanwhile, let L be the splitting field of $f_1 \cdot f_2$. Then, L contains all roots of $f_1 f_2$, so $K_1K_2 \subseteq L$. And L is splitting field of $f_1 f_2$, so $L \subseteq K_1K_2$, so $L = K_1K_2$ is splitting field.

Let $\alpha \in K_1 \cap K_2 \setminus F$ be given. Then, there is minimal irreducible polynomial $f(x)$ over F having α as a root. Since $\alpha \in K_1$ and $\alpha \in K_2$, the $f(x)$ splits in both K_1 and K_2, so for $\{\alpha_1, \cdots, \alpha_n\}$, the set of all roots of f, $F(\alpha_1, \cdots, \alpha_n) \subseteq K_1 \cap K_2$, so f splits over $K_1 \cap K_2$.
Since α was arbitrary, the $K_1 \cap K_2$ is splitting field

Def. A field K/F is called separable if:
every element of K is the root of a separable polynomial over F.

「Summit. 13,5

「Prop. Let F be any field, and $d, n \in \mathbb{N}$. Then

d divides n if and only if $x^d - 1$ divides $x^n - 1$.

Proof. Suppose that $n = dk$ for some $k \geq 2$. (If $k=1$, we are done)

$$x^n - 1 = x^{dk} - 1 = (x^d)^k - 1 = (x^d - 1)(x^{d(k-1)} + x^{d(k-2)} + \cdots + 1)$$

So, $x^d - 1$ divides $x^n - 1$. Conversely, suppose that $x^d - 1$ divides $x^n - 1$.
By the division algorithm, $n = qd + r$ for some $q, r \in \mathbb{Z}$ and $0 \leq r < d$.

So, $x^n - 1 = x^{qd+r} - x^r + x^r - 1 = x^r \cdot (x^{qd} - 1) + x^r - 1$. ($q \neq 0$, since $n \geq d$)

Since $x^{qd} - 1 = (x^d - 1)(x^{d(q-1)} + \cdots + x + 1)$ is divided by $x^d - 1$,
by the assumption, $x^r - 1$ is divided by $x^d - 1$. But, since $r < d$,
the remainder r must be zero, so $n = qd$. 証

「Prop. $\mathbb{F}_{p^d} \subseteq \mathbb{F}_{p^n}$ if and only if d divides n.

proof. Since

$$\mathbb{F}_{p^k} = \{ x \in \overline{\mathbb{F}_p} \mid x^{p^k} - x = 0 \}$$

and the fact that

$$\mathbb{F}_{p^d} \subseteq \mathbb{F}_{p^n} \iff x^{p^d - 1} - 1 \text{ divides } x^{p^n - 1} - 1$$
$$\iff p^d - 1 \text{ divides } p^n - 1$$
$$\iff d \text{ divides } n$$

So proved.